

(An Autonomous Institution affiliated to VTU, Belagavi)

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Fifth Semester B.E. (Robotics and AI) (Credit System) Degree Examinations

Supplementary Examinations – December 2025

21RI503 – KINEMATICS AND DYNAMICS OF ROBOT

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I

Marks	BT*	CO*	PC
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1. a) Obtain the direct kinematic solution for the robot configuration shown in **Fig.Q1(a)(a)**, assuming home position, reference coordinate system and hand coordinate system as shown in **Fig. Q1(a)(b)**. Determine the position and orientation of the hand coordinate system if the joint variables are $\theta_1 = \theta_2 = \theta_4 = 0$ and $d_3 = 100$ units, $l_1 = l_4 = 100$ units.

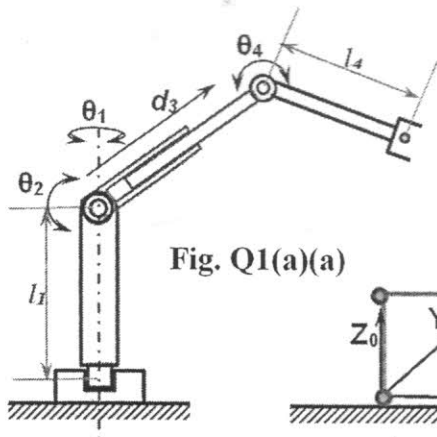


Fig. Q1(a)(a)

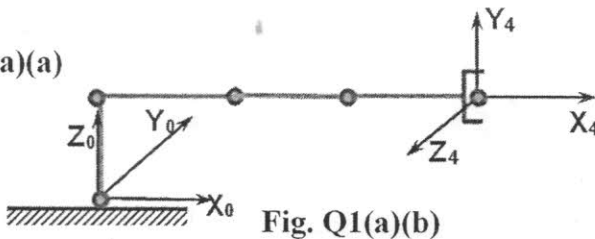


Fig. Q1(a)(b)

- b) Explain the procedure to be followed to obtain the inverse kinematics solution for a given robot.
2. a) For the 3R robot shown in the figure obtain the direct kinematics solution assuming the home position as shown. Also assume the base and end effector coordinate system as shown in the figure. Also find the position and orientation of the end effector if $\theta_1 = \theta_2 = \theta_3 = 0$; and $l_1 = 50$, $l_2 = l_3 = 100$.

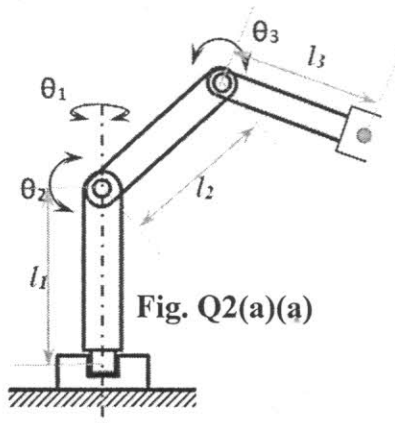


Fig. Q2(a)(a)

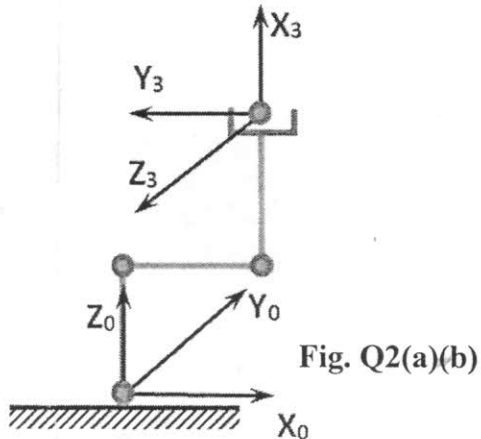


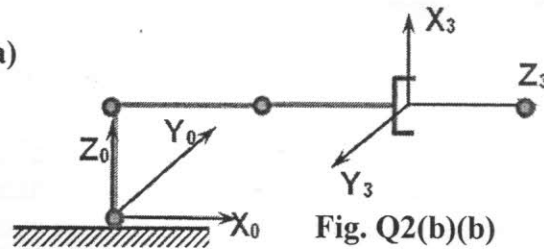
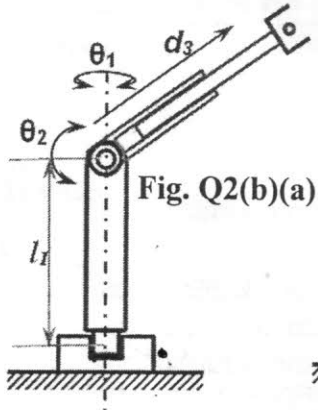
Fig. Q2(a)(b)

- b) Obtain the inverse kinematics solution for the robot configuration shown in **Fig.Q2(b)(a)**, assuming home position, reference coordinate system and hand coordinate system as shown in **Fig. Q2(b)(b)** and the transformation of joint coordinate systems 0T_1 , 1T_2 , 2T_3 and the direct kinematic solution 0T_3 is given by.

$${}^0T_1 = \begin{bmatrix} \cos \theta_1 & 0 & \sin \theta_1 & 0 \\ \sin \theta_1 & 0 & -\cos \theta_1 & 0 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^2T_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

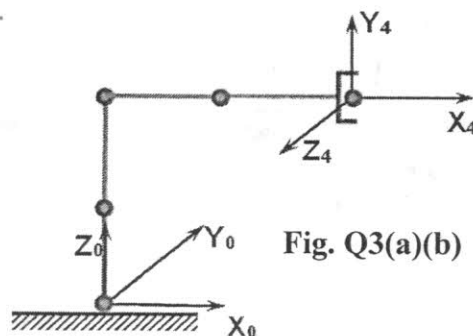
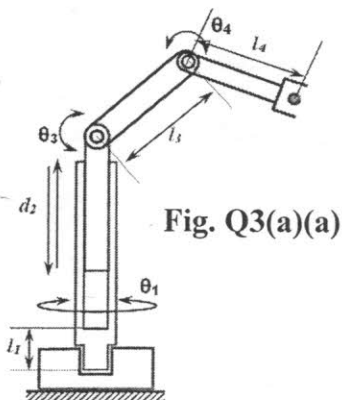
$${}^1T_2 = \begin{bmatrix} \sin \theta_2 & 0 & \cos \theta_2 & 0 \\ \cos \theta_2 & 0 & \sin \theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_3 = \begin{bmatrix} c\theta_1 s\theta_2 & s\theta_1 & c\theta_1 c\theta_2 & c\theta_1 d_3 c\theta_2 \\ s\theta_1 s\theta_2 & -c\theta_1 & s\theta_1 c\theta_2 & s\theta_1 d_3 c\theta_2 \\ c\theta_2 & 0 & s\theta_2 & d_3 s\theta_2 + l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



10 L3 2 1

3. a) Obtain the direct kinematic solution for the robot configuration shown in **Fig.Q3(a)(a)**, assuming home position, reference coordinate system and hand coordinate system as shown in **Fig. Q3(a)(b)**. Determine the position and orientation of the hand coordinate system if the joint variables are $\theta_1 = \theta_3 = \theta_4 = 0$ and $d_2 = 100$ units, $l_3 = l_4 = 100$ units, $l_1 = 100$ units.



10 L3 1 1

- b) For the 3DOF articulated arm Shown in Fig. Q3(a), obtain the inverse kinematic solution if the direct kinematic solution is given by the transformation matrix T.

$${}^0T_3 = \begin{bmatrix} C_1 C_{23} & -C_1 S_{23} & S_1 & C_1 (L_3 C_{23} + L_2 C_2) \\ S_1 C_{23} & -S_1 S_{23} & -C_1 & S_1 (L_3 C_{23} + L_2 C_2) \\ S_{23} & C_{23} & 0 & L_3 S_{23} + L_2 S_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

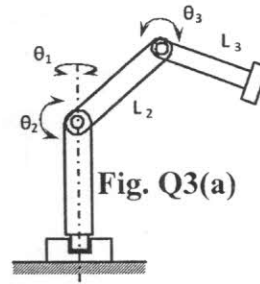


Fig. Q3(a)

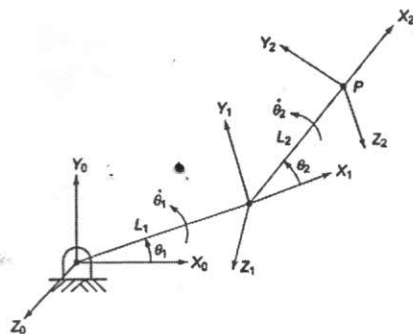
The transformation of joint coordinate systems ${}^0T_1, {}^1T_2, {}^2T_3$ is given by

$${}^0T_1 = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; {}^1T_2 = \begin{bmatrix} C_2 & -S_2 & 0 & L_2 C_2 \\ S_2 & C_2 & 0 & L_2 S_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix};$$

$${}^2T_3 = \begin{bmatrix} C_3 & -S_3 & 0 & L_3 C_3 \\ S_3 & C_3 & 0 & L_3 S_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Unit – II

4. a) Derive the equation for the linear velocity and angular velocity of a link-i
b) Determine the singularities and joint velocities of 2-DOF RR Planar manipulator. Assume $L_1 = L_2 = 1$



$${}^0T_1 = \begin{bmatrix} C_1 & -S_1 & 0 & C_1 \\ S_1 & C_1 & 0 & S_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; {}^1T_2 = \begin{bmatrix} C_2 & -S_2 & 0 & C_2 \\ S_2 & C_2 & 0 & S_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_2 = \begin{bmatrix} C_1 C_2 - S_1 S_2 & -C_1 S_2 - S_1 C_2 & 0 & C_1 + C_1 C_2 - S_1 S_2 \\ C_1 S_2 + S_1 C_2 & C_1 C_2 - S_1 S_2 & 0 & S_1 + C_1 S_2 + S_1 C_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5. a) Obtain the Jacobian matrix for the robot configuration shown in Fig.Q2(b)(a), assuming home position, reference coordinate system and hand coordinate system as shown in Fig. Q2(b)(b) and the transformation of joint coordinate systems is given by ${}^0T_1, {}^1T_2, {}^2T_3$ and the direct kinematic solution is represented by transformation matrix 0T_3 given in Question number 2(b).

- b) Determine the singularities of RPY Wrist, the transformation of joint coordinate systems 0T_1 , 1T_2 , 2T_3 and the direct kinematic solution 0T_3 is given

$${}^0T_1 = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^1T_2 = \begin{bmatrix} -S_2 & 0 & C_2 & 0 \\ C_2 & 0 & S_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^2T_3 = \begin{bmatrix} C_3 & -S_3 & 0 & 0 \\ S_3 & C_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_3 = \begin{bmatrix} -C_1S_2C_3 + S_1S_3 & C_1S_2S_3 + S_1C_3 & C_1C_2 & 0 \\ -S_1S_2C_3 - C_1S_3 & S_1S_2S_3 - C_1C_3 & S_1C_2 & 0 \\ C_1C_3 & -C_2S_3 & S_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

6. a) Derive the equation for obtaining the linear velocity due to angular motion
 b) What is Jacobian Singularities? What are the significance of Jacobian Singularities? What is the meaning of Boundary Singularities and Interior Singularities?

10	L3	4	1
10	L2	3	1
10	L1	4	1

Unit – III

7. a) Obtain the inertia coefficients for the 2-DOF planar manipulator as shown in the Fig. Q2. Assume both links have equal lengths ($L_1 = L_2 = L$) and have equal masses ($m_1 = m_2 = m$). Assume further that the links are slender members with a uniform mass distribution, that is, the center of mass of each link is located at the midpoint of the link. Assume also that the inertia tensors I_1 and I_2 with respect to frame $\{i\}$, $i = 1, 2$ as:

$$I_1 = I_2 = \begin{bmatrix} mL^2/3 & 0 & 0 & -mL/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -mL/2 & 0 & 0 & m \end{bmatrix}$$

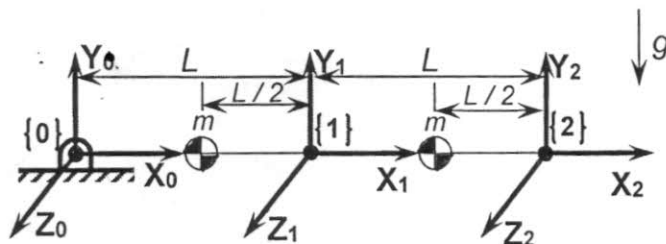


Fig. Q7(a)

- b) One of the joints of a SCARA manipulator is required to move from 0° to 120° in 5 seconds. Find the cubic polynomial to generate the smooth trajectory for the joint. What is the maximum velocity and acceleration for this joint trajectory? Plot the displacement, Velocity and acceleration profile with respect to time.
8. a) Write the various terms in the dynamic equation of motion of 3-DOF manipulator using Lagrange-Euler formulations.
 b) Derive the expression for the position, velocity and acceleration of a robot joint motion assuming cubic polynomial trajectory for the joint

10	L3	5
10	L3	5
10	L1	5
10	L2	5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
